

Elec-S1 · Part 1

Progress Tests 1.A → 3.C

Foundations of DC circuit theory

One short test per lecture section, in the format of Test 1

Semester 1 · Academic Year 2026–2027

How to use this booklet

1. Study a section of the lecture notes, then take its progress test *in real conditions*: timer on, no notes, no calculator, answers only inside the frames.
2. Mark yourself with the answer-key booklet. Each circled number in the margin is one marking item; the scheme below each exercise says exactly what earns it.
3. Convert your score: mark out of 20 = score $\times$ 20/14. Write it in the tracker (next page). Below 14/20, re-read the section and retake the test a few days later.
4. When all nine tests are done, take the **End-of-Unit Test** (45 minutes, 50 marks, sections 1.A to 3.C).

Contents

Test	Lecture section	Notes, pages	Duration	Marks
1.A	From potential to electric voltage	2–7	7 min	14
1.B	From charge to electric current	7–11	7 min	14
1.C	Circuits and two-terminal components	11–14	7 min	14
2.A	Characteristic curve, evidence for Ohm's law	14–19	10 min	14
2.B	The ohmic conductor and Ohm's law	19–21	7 min	14
2.C	Pouillet's law	21–22	7 min	14
3.A	Kirchhoff's laws (combinations, divider, Millman)	22–29	10 min	14
3.B	Thévenin's and Norton's equivalence theorems	29–33	10 min	14
3.C	Operating point of a circuit	34–38	10 min	14

Every test mirrors Test 1 (21 September 2026): a unit conversion, a definition with its equation and SI units, tick-box questions, and a short calculation with the data framed. The numbers are chosen so that everything can be done without a calculator.

Progress tracker

Test	Date	/14	/20	Retake date	/14	To revise
1.A						
1.B						
1.C						
2.A						
2.B						
2.C						
3.A						
3.B						
3.C						
End of unit		/50		/20 = score $\times$ 20/50		

Before every test, remember Test 1

- Letters first, numbers second: isolate the unknown literally, *then* substitute.
- Every number carries its unit, and every unit symbol is exact ($k \neq K$, $m \neq M$).
- Powers of ten separately from the mantissas; check the order of magnitude at the end.
- Nodes = electrically distinct nodes; each plain wire segment is a two-terminal component.

Progress Test 1.A — Elec-S1

1.A · From potential to electric voltage
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion (1 mark)

Convert the voltage $u = 0.0258\text{ MV}$ into millivolts. Give the exact result in scientific notation, without rounding.

[Empty grid for answer]

Marking

①
 0 1
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Exercise 2. Electric voltage — definition (5 marks)

Define the electric voltage u_{AB} between two points A and B of a circuit. Write down its expression in terms of the electric potentials, deduce the relation between u_{AB} and u_{BA} , and state the SI unit of voltage (name and symbol).

[Empty grid for answer]

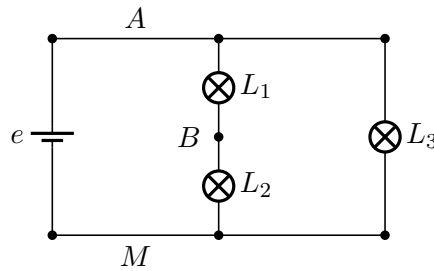
Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
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Exercise 3. Potentials in a circuit

(3 marks)

The circuit below contains an ideal voltage source of electromotive force $e = 6.0 \text{ V}$ and three lamps. The wires are ideal. The wire M , connected to the $-$ terminal of the source, is taken as the reference of potentials: $V_M = 0$.



How many distinct electric potentials are there in this circuit?

- ☐ 2 ☐ 3 ☐ 4 ☐ 7

The voltage across L_2 is $u_{BM} = 2.0 \text{ V}$. The voltage u_{AB} across L_1 is:

- ☐ 2.0 V ☐ 4.0 V ☐ 6.0 V ☐ 8.0 V

A voltmeter has its V terminal on B and its COM terminal on A . It displays:

- ☐ +4.0 V ☐ -4.0 V ☐ +2.0 V ☐ -6.0 V

Exercise 4. Potentials and voltages

(5 marks)

In a circuit, the ground M is taken as the reference of potentials ($V_M = 0$). Three points A , B and C have the potentials given below. Express, then compute, the voltages u_{AB} , u_{BC} and u_{CA} . Compute their sum and comment.

$V_A = 12.0\text{ V}$	$V_B = 4.5\text{ V}$	$V_C = -3.0\text{ V}$
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Marking

①	
0	1

②	
0	1

③	
0	1

④	
0	1

⑤	
0	1

— End of the test —

Progress Test 1.B — Elec-S1

1.B · From charge to electric current
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion (1 mark)

The capacity of a phone battery is $Q = 4000\text{ mA h}$, i.e. the charge it delivers when a current of 4000 mA flows for one hour. Express Q in coulombs. Give the exact result.

[Empty grid for Exercise 1 answer]

Marking

①
 0 1
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Exercise 2. Microscopic expression of the current (5 marks)

A conductor of cross-sectional area S contains n mobile charge carriers per unit volume, each of charge q , all moving with the drift velocity v along the axis of the conductor. Express the charge dq that crosses S during dt (explain your reasoning), deduce the current i , and check that your result is dimensionally consistent, stating the SI unit of n .

[Empty grid for Exercise 2 answer]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Current and ammeters

(3 marks)

In a copper wire carrying a current, the conduction electrons move:

- ☐ in the conventional direction of the current;
- ☐ in the direction opposite to the conventional direction of the current;
- ☐ at a speed close to that of the electric signal;
- ☐ only when the wire is heated.

An ammeter is connected:

- ☐ in parallel across the component, and has a very large internal resistance;
- ☐ in series in the branch, and has a very small internal resistance;
- ☐ in series in the branch, and has a very large internal resistance;
- ☐ in parallel across the component, and has a very small internal resistance.

Which of these units is one of the seven SI base units?

- ☐ coulomb (C) ☐ volt (V) ☐ ampere (A) ☐ ohm (Ω)

Exercise 4. Counting electrons

(5 marks)

A steady current i flows through a wire during a time Δt . Determine the number N of electrons that cross a cross-section of the wire during Δt .

$i = 3.2 \text{ mA}$	$\Delta t = 5.0 \text{ s}$	$e = 1.6 \times 10^{-19} \text{ C}$
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Marking

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0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

— End of the test —

Semester 1 · Academic Year 2026–2027

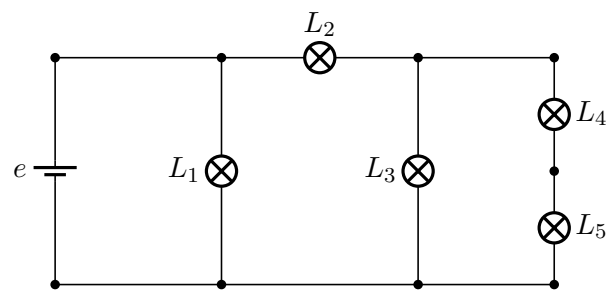
①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Reading a circuit diagram

(4 marks)



Base your answers on the diagram exactly as drawn, even though the same circuit could be drawn with fewer wires.

How many nodes?

- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 9

How many branches?

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 11

How many two-terminal components, connecting wires included?

- ☐ 6
- ☐ 9
- ☐ 11
- ☐ 14

How many independent meshes (independent mesh equations)?

- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Exercise 4. Energy received by a phone

(4 marks)

A phone is charged in the DC regime: the charger applies $u = 5.0\text{ V}$ and delivers $i = 2.0\text{ A}$ during $\Delta t = 1\text{ h } 30\text{ min}$. Compute the energy received by the phone, in joules, then in watt-hours.

$u = 5.0\text{ V}$	$i = 2.0\text{ A}$	$\Delta t = 1\text{ h } 30\text{ min}$

Marking

①

01

②

01

③

01

④

01

— End of the test —

Progress Test 2.A — Elec-S1

2.A · Characteristic curve and experimental evidence for Ohm’s law
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Reading a slope (1 mark)

The characteristic $i = f(u)$ of a resistor is a straight line through the origin, of slope 20 mA V^{−1}. Give its resistance R .

[Empty grid for Exercise 1]

Marking

①
 0 1
 [] []

Exercise 2. Characteristic of a component — definition (5 marks)

Define the characteristic of a two-terminal component. Which convention is used to plot the characteristic of a passive component? Draw it and give the sign of the power received. What is the shape of the characteristic of an ohmic conductor?

[Empty grid for Exercise 2]

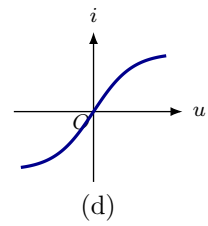
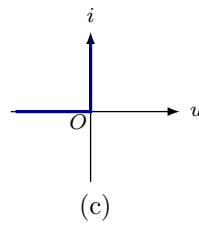
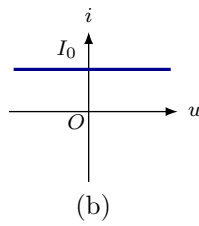
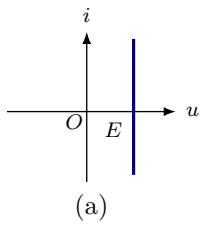
Marking

①
 0 1 2
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 ②
 0 1 2
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 ③
 0 1
 [] []

Exercise 3. Identifying characteristics

(3 marks)

The four characteristics below are plotted in the (u, i) plane.



Characteristic (a) is that of:

- ☐ an ohmic conductor
 - ☐ an ideal voltage source
 - ☐ an ideal current source
 - ☐ a lamp

Characteristic (c) is that of:

- ☐ a Zener diode
 - ☐ an ideal diode
 - ☐ a lamp
 - ☐ an ideal current source

Which statement is correct?

- ☐ Every characteristic is a straight line through the origin.
- ☐ A straight-line characteristic through the origin is the signature of an ohmic conductor.
- ☐ The characteristic of a lamp is a straight line through the origin.
- ☐ A characteristic does not depend on how u and i are oriented.

Exercise 4. Estimating a resistance from measurements

(5 marks)

The characteristic of a resistor was measured in the receiver convention. The measurements at low current are the noisiest.

i_k (mA)	1.0	2.0	4.0
u_k (V)	0.70	0.90	2.00

(a) Compute the naive estimate $R_{\text{naive}} = \frac{1}{N} \sum_k u_k / i_k$. (b) Compute the least-squares estimate $\hat{R} = \sum_k u_k i_k / \sum_k i_k^2$. (c) Which value should be kept? Justify.

This image shows a full page of blank graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

Marking

1

01

2

01

3

01

4

01

5

01

— End of the test —

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(3 marks)

☐ a few m Ω ☐ a few Ω ☐ a few k Ω ☐ 1 to 10 M Ω

- ☐ leave the circuit powered, so that a current flows through the resistor;
- ☐ switch the power supply off, but leave the resistor connected;
- ☐ disconnect the power supply and at least one terminal of the resistor;
- ☐ connect the ohmmeter in series with the resistor.

$$\square \quad Ri \qquad \square \quad R^2i \qquad \square \quad Ri^2 \qquad \square \quad u^2R$$

(5 marks)

$$u = 12 \text{ V} \qquad p = 72 \text{ W} \qquad \Delta t = 10 \text{ min}$$

1

01

2

01

3

01

4

01

5

01

Unofficial practice material written from the Elec-SI Part 1 lecture notes · not an official examination paper

Semester 1 · Academic Year 2026–2027

Progress Test 3.A — Elec-S1
 3.A · Kirchhoff’s laws (combinations, divider, Millman)
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Equivalent resistance (1 mark)

Give the equivalent resistance of $R_1 = 6.0\text{ k}\Omega$ and $R_2 = 3.0\text{ k}\Omega$ connected in parallel.

[Empty grid for answer]

Marking

①
 0 1
 [] []

Exercise 2. Kirchhoff’s laws (5 marks)

State Kirchhoff’s node law and mesh law. For each law, give the fundamental principle it expresses.

[Empty grid for answer]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Quick circuit laws

(3 marks)

$R_1 = 1.0\text{ k}\Omega$ and $R_2 = 3.0\text{ k}\Omega$ are in series under $u = 12\text{ V}$. The voltage u_2 across R_2 is:

- ☐ 3.0 V ☐ 4.0 V ☐ 9.0 V ☐ 12 V

A circuit has $N = 3$ electrically distinct nodes and $B = 5$ branches. The number of independent mesh equations is:

- ☐ 2 ☐ 3 ☐ 5 ☐ 8

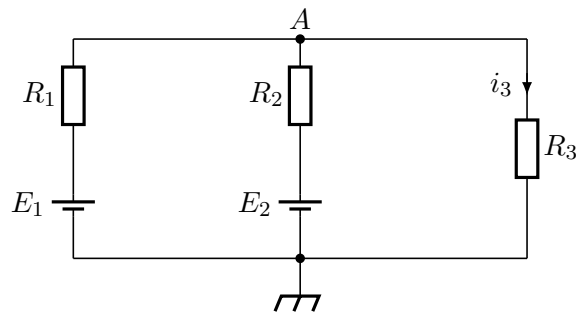
Three identical $30\ \Omega$ resistors in parallel are equivalent to:

- ☐ $90\,\Omega$ ☐ $30\,\Omega$ ☐ $10\,\Omega$ ☐ $3.3\,\Omega$

Exercise 4. Millman's theorem

(5 marks)

In the circuit below, the ground (bottom wire) is the reference of potentials. Using Millman's theorem, determine the potential V_A of node A . Deduce the current i_3 that flows through R_3 towards the ground.



$E_1 = 12\text{ V}$	$R_1 = 2.0\ \Omega$	$E_2 = 6.0\text{ V}$	$R_2 = 3.0\ \Omega$	$R_3 = 6.0\ \Omega$
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[illegible]

Marking

1

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— End of the test —

Progress Test 3.B — Elec-S1
 3.B · Thévenin’s and Norton’s equivalence theorems
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. From Norton to Thévenin (1 mark)

A Norton model has $I_{No} = 2.0\text{ mA}$ and $R_{No} = 4.7\text{ k}\Omega$. Give the electromotive force E_{Th} of the equivalent Thévenin model.

Marking

Exercise 2. Thévenin’s theorem (5 marks)

State Thévenin’s theorem for a network made of ideal sources and ohmic conductors, seen from its terminals A and B (draw the model). Explain how E_{Th} and R_{Th} are determined.

Marking

(3 marks)

- ☐ an open circuit
- ☐ a resistor R_{Th}
- ☐ a short circuit (a wire)
- ☐ an ideal current source

☐ $I_{\text{No}} = 36 \text{ A}, R_{\text{No}} = 3.0 \Omega$

☐ $I_{\text{No}} = 4.0 \text{ A}, R_{\text{No}} = 3.0 \Omega$

☐ $I_{\text{No}} = 4.0 \text{ A}, R_{\text{No}} = 0.33 \Omega$

☐ $I_{\text{No}} = 0.25 \text{ A}, R_{\text{No}} = 3.0 \Omega$

- ☐ a diode;
- ☐ only ideal sources and ohmic conductors;
- ☐ a lamp;
- ☐ a motor.

(5 marks)

The circuit diagram shows a DC voltage source E connected in series with a resistor R_1 . After R_1 , the circuit splits into two parallel branches: one containing resistor R_2 and the other containing resistor R_3 . Both branches recombine, and the output terminals are labeled A and B . The voltage across the output terminals is denoted by u .

$E = 12\text{ V}$	$R_1 = 6.0\text{ k}\Omega$	$R_2 = 3.0\text{ k}\Omega$	$R_3 = 1.0\text{ k}\Omega$
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[illegible]

1

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2

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3

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4

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5

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Unofficial practice material written from the Elec-S1 Part 1 lecture notes · not an official examination paper

Progress Test 3.C — Elec-S1

3.C · Operating point of a circuit
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Maximum power (1 mark)

A source has $E_{Th} = 12\text{ V}$ and $R_{Th} = 6.0\,\Omega$. Give the maximum power it can deliver to a resistive load.

[Empty grid for answer]

Marking

①
 0 1
 [] []

Exercise 2. The operating point (5 marks)

A linear network, described by its Thévenin model (E_{Th}, R_{Th}), feeds a load resistor R_c . Define the operating point. Write down the characteristic of the source and that of the load, and deduce the analytical expressions of u^* and i^* .

[Empty grid for answer]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

